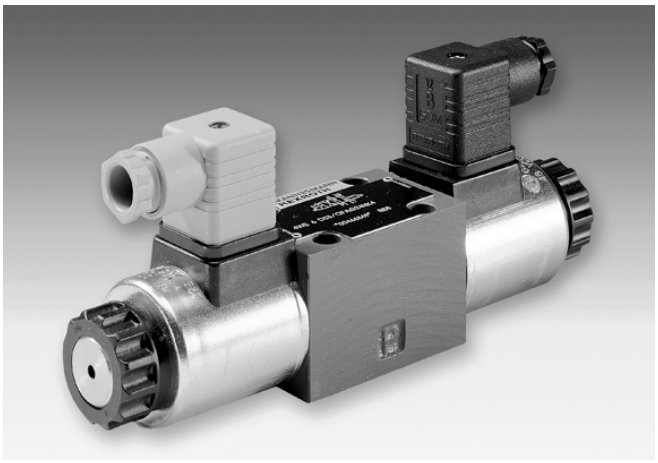
RE 23 178/09.99

Replaces: 08.96

4/3, 4/2 and 3/2 directional valves with wet pin DC or AC solenoids, Type WE 6 ../.E

Nominal size 6
Series 6X
Maximum operating pressure 350 bar
Maximum flow 80 L/min



HAD5909

Type 4WE 6 E6X/EG24N9K4 with plug-in connector

Overview of contents

Contents	Page
Features	1
Ordering details	2, 3
Symbols	2
Function, section	3
Technical data	4
Performance limits	5
Characteristic curves	6
Preferred types	6
Unit dimensions	7, 8
Available spare parts and seals	9

Features

- Direct solenoid actuated directional spool valve high performance version
- Porting pattern to DIN 24 340 form A, ISO 4401 and CETOP–RP 121 H, For subplates see catalogue sheet RE 45 052 (separate order)
- Wet pin DC or AC solenoids with removable coil
- Solenoid coil can be rotated through 90°
- It is not necessary to open the pressure tight chamber when changing the coil
- Electrical connections either as individual or central connections
- Hand override, optional
- Soft switching version, see RE 23 183
- Inductive limit switch (contact or inductive), see RE 24 830

Ordering details

2	3	4	6	7	9	10	11	12	15	19	22	23
	WE	6		6X	/	E			/			*

3 service ports = **3**
4 service ports = **4**

Nominal size 6 = **6**

Symbol e.g. C, E, EA, EB etc.
for possible designs see below

Series 60 to 69 = **6X**
(60 to 69: unchanged installation and connection dimensions)

Spring return = **No code**
Without spring return = **O**
Without spring return but with detent = **OF**

High power solenoid = **E**
Wet pin (oil immersed) with removable coil

24 V DC = **G24**
230 V AC 50/60 Hz = **W230**
205 V DC = **G205** [2]
For the ordering details of other voltages and frequencies see page 4

With protected hand override (standard) = **N9**
With hand override = **N**
Without hand override = **No code**

Further details in clear text

No code = NBR seals
V = FKM seals
(other seals on request)

⚠ Attention!
The compatibility of the seals and pressure fluid has to be taken into account!

No code = Without cartridge throttle
B08 = Throttle Ø 0.8 mm
B10 = Throttle Ø 1.0 mm
B12 = Throttle Ø 1.2 mm
Used where the flow is > than the performance limit of valve, active in the P line

Electrical connections

Individual connections

K4 [1] **=** Individual connection; with component plug DIN 43 650-AM2, without plug-in connector

Central connections

DL = Terminal box with cable connector, with indicator light
DKL [3] **=** Central connection on cover with indicator light (without angled plug-in connector)

AC supply voltage (permissible voltage tolerance ± 10%)	**Nominal voltage of DC solenoids when used with an AC supply**	**Ordering details**
110 V - 50/60 Hz	96 V	**G96**
230 V - 50/60 Hz	205 V	**G205**

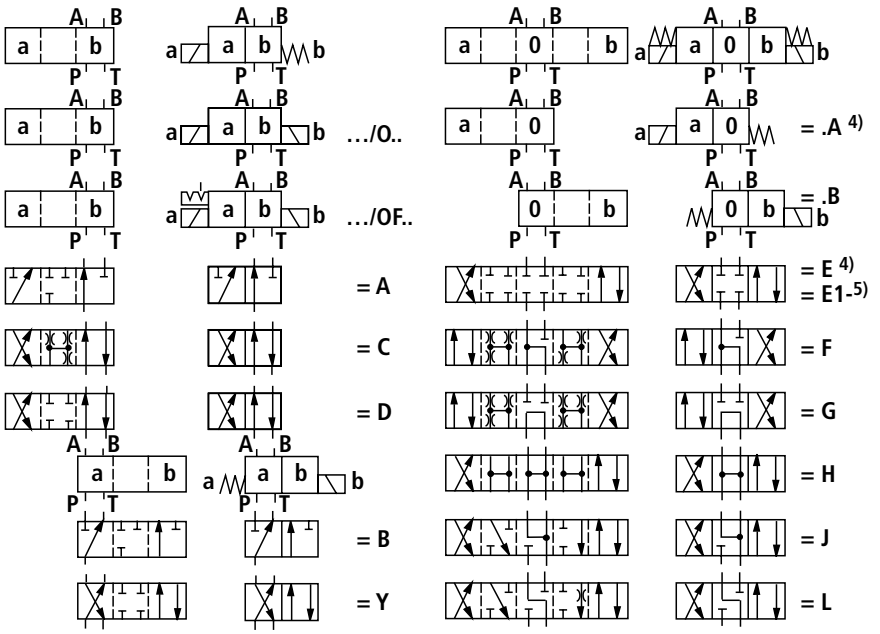
Preferred types, see page 6, are readily available!

1) Plug-in connectors must be ordered separately (see page 3).

2) When connecting to an AC supply a DC solenoid **must** be used which is controlled via a rectifier (see table left).
With an individual connection a large plug-in connector with built-in rectifier can be used (separate order, see page 3).

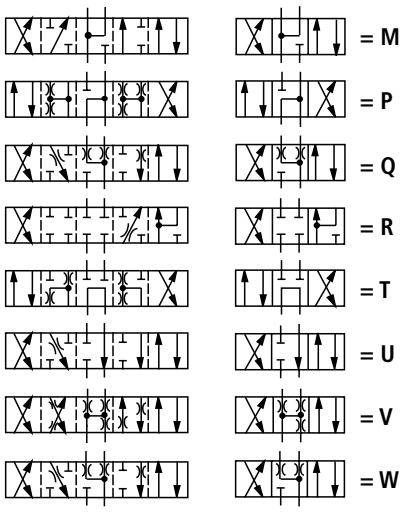
3) Angled plug-in connector (Material no. 00005538) must be ordered separately.

Symbols

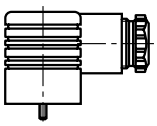
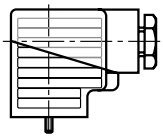


4) **Example:** Spool E with switching position "a" ordering details..**EA**..

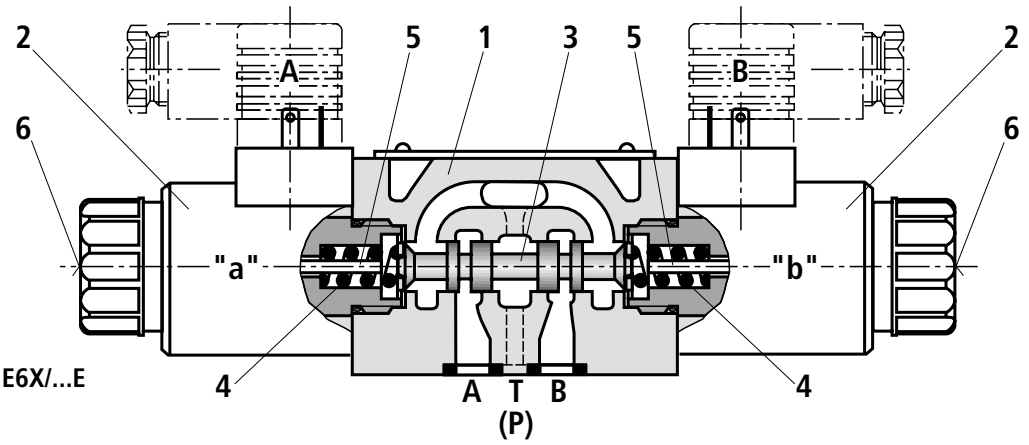
5) Symbol E1–: P – A/B pre-opening
Attention: Regarding pressure intensification with differential cylinders!



Ordering details: plug-in connectors to DIN 43 650 A and ISO 4400 for component plug "K4"

For further plug-in connectors see RE 08 006					
		Material no.			
Valve side	**Colour**	Without circuitry	With indicator light 12 … 240 V	With rectifier 12 … 240 V	With indicator light and Z-diode protective circuit 24 V
a	grey	**00074683**	–	–	–
b	black	**00074684**	–	–	–
a/b	black	–	**00057292**	**00313933**	**00310995**

Function, section



Type 4WE 6 E6X/...E

Type WE directional valves are solenoid operated directional spool valves. They control the start, stop and direction of flow.

Essentially the directional control valves consist of housing (1), one or two solenoids (2), the control spool (3), and one or two return springs (4).

In the de-energised condition the control spool (3) is held in the neutral or initial position by means of return springs (4) (except for impulse spools). The control spool (3) is actuated via wet pin solenoids (2).

To guarantee satisfactory operation care should be taken to ensure that the solenoid pressure chamber is filled with oil.

The force of the solenoids (2) acts via the plunger (5) on the control spool (3) and pushes this from its neutral position to the required end position. This gives free-flow from P to A and B to T or P to B and A to T.

When solenoid (2) is de-energised, the control spool (3) is returned to its neutral position by means of the return springs (4).

An optional hand override (6), allows movement of the control spool (3) without energising the solenoid.

Type 4WE 6.. 6X/O... (only possible for symbols A, C and D)

This design is for directional control valves with 2 switched positions and 2 solenoids without detent. There is no definable switched position when the solenoids are de-energised.

Type 4WE 6.. 6X/OF... (impulse spool, only for symbols A, C and D)

This design is for directional control valves with 2 switched positions, 2 solenoids and a detent. Both switched positions are thus fixed alternately and there is no need to continually energise the solenoid.

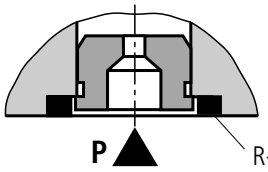
Note:

Pressure peaks in the tank line to two or more valves can, particularly with valves with detents, lead to unintended spool movements! It is therefore, recommended that a separate tank line is used.

Cartridge throttle (type 4WE 6..6X/…/**B**..)

If, due to particular operating conditions during the switching sequences, flows can occur which are larger than the valve performance curves allow, then it will be necessary to fit a cartridge throttle.

This is inserted in the P channel of the directional control valve.



R-ring 9.81 x 1.5 x 1.78

Technical data (for applications outside these parameters, please consult us!)

General

Installation			optional
Max. ambient temperature		°C	50
Weight	Valve with 1 solenoid	kg	1.45
	Valve with 2 solenoids	kg	1.95

Hydraulic

Max. operating pressure	Ports A, B, P	bar	350
	Port T	bar	210 (=) ; 160 (~) with symbols A and B, port T must be used as a drain port if the operating pressure is above the permitted tank pressure.
Max. flow		L/min	80 (=); 60 (~)
Flow cross-section (switched position 0):			
	For symbol Q	mm^2	approx. 6 % of the nominal cross-section
	For symbol W	mm^2	approx. 3 % of the nominal cross-section
Pressure fluid			mineral oil (HL, HLP) to DIN 51 524 [1]; fast bio degradable pressure fluids to VDMA 24 568 (also see RE 90 221); HETG (rape seed oil) [1]; HEPG (polyglycol) [2]; HEES (synthetic ester) [2]; other pressure fluids on request
Pressure fluid temperature range		°C	– 30 to + 80 (NBR seals)
			– 20 to + 80 (FKM seals)
Viscosity range		mm^2/s	2.8 to 500
Degree of contamination			maximum permissible degree of contamination of the pressure fluid is to NAS 1638 class 9. We, therefore, recommend a filter with a minimum retention rate of $\beta_{10} \geq 75$.

Electrical

Voltage type			DC	AC 50/60 Hz
Available voltages [3] (for ordering details af AC solenoids see below)		V	12, 24, 96, 205	110, 230
Voltage tolerance (nominal voltage)		%	±10	
Power consumption		W	30	–
Holding power		VA	–	50
Switch-on power		VA	–	220
Duty			continuous	continuous
Switching time to ISO 6403	ON	ms	25 to 45	10 to 20
	OFF	ms	10 to 25	15 to 40
Switching frequency		cycles/h	up to 15000	up to 7200
Protection to DIN 40 050			IP 65	IP 65
Max. coil temperature [4]		°C	150	180

[1] Suitable for NBR **and** FKM seals

[2] **Only** suitable for FKM seals

[3] Special voltages on request

[4] Due to the occuring surface temperatures of the solenoid coils, the European standards EN563 and EN982 must be taken into account!

With electrical connections the protective conductor (PE ⏚) must be connected according to the relevant regulations.

Note:
AC solenoids may be used for 2 or 3 types of supply;
e.g. solenoid type **W110** for:
110 V, 50 Hz
110 V, 60 Hz
120 V, 60 Hz

Ordering details	
W110	110 V, 50 Hz 110 V, 60 Hz 120 V, 60 Hz
W230	230 V, 50 Hz 230 V, 60 Hz

Performance limits (measured at $\nu = 41$ mm²/s and $\vartheta = 50$ °C)

⚠ Attention!

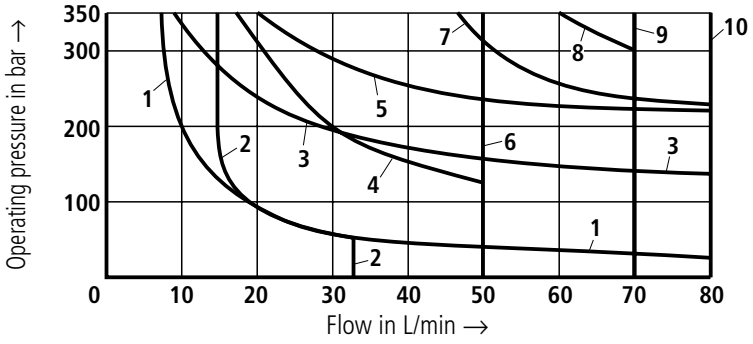
The given switching power limits are for applications with two flow directions (e. g. from P to A and simultaneous return flow from B to T).

Due to the flow forces active within the valves the permissible switching power limit may be significantly less if there is only one direction of flow (e. g. from P to A and port B blocked)!

(Please consult us for applications of this kind.)

The switching power limits were measured with the solenoids at operating temperature, 10% under voltage and without tank back pressure.

DC solenoid		AC solenoid – 50 Hz		AC solenoid – 60Hz	
Char. curve	**Symbol**	**Char. curve**	**Symbol**	**Char. curve**	**Symbol**
1	A, B[1)]	**11**	A, B[1)]	**19**	A, B[1)]
2	V	**12**	V	**20**	V
3	A, B	**13**	A, B	**21**	A, B
4	F, P	**14**	F, P	**22**	F, P
5	J	**15**	G, T	**23**	G, T
6	G, H, T	**16**	H	**24**	J, L, U
7	A/O, A/OF, L, U	**17**	A/O, A/OF, C/O, C/OF	**25**	A/O, A/OF, Q, W
8	C, D, Y		D/O, D/OF, E, E1–[2)], J, L	**26**	C, D, Y
9	M		M, Q, R[3)], U, W	**27**	H
10	E,E1–[2)], R[3)], C/O, C/OF D/O, D/OF, Q, W	**18**	C, D, Y	**28**	C/O, C/OF, D/O, D/OF, E, E1–[2)], M, R[3)]



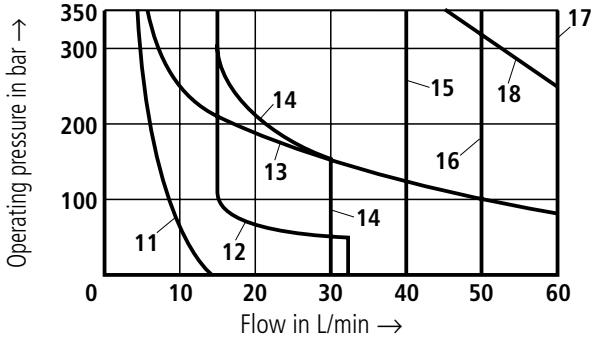
[1)] With hand override

[2)] P – A/B pre-opening

[3)] Return flow from actuator to tank

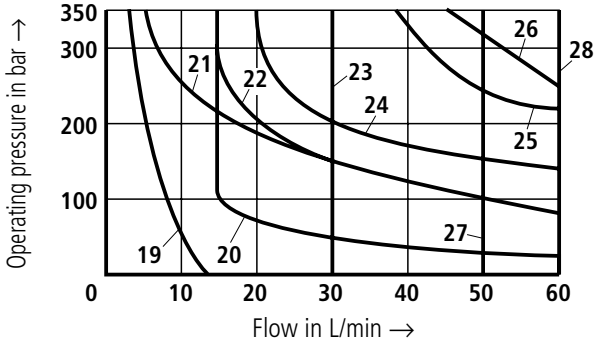
DC solenoid	
Char. curve	**Solenoid voltage**
1 to 10	12, 24, 96, 205 V

(other voltages on request)



AC solenoid		
Char. curve	**Solenoid voltage**	
11 to 18	W110	110 V, 50 Hz 120 V, 60 Hz
	W230	230 V, 50 Hz

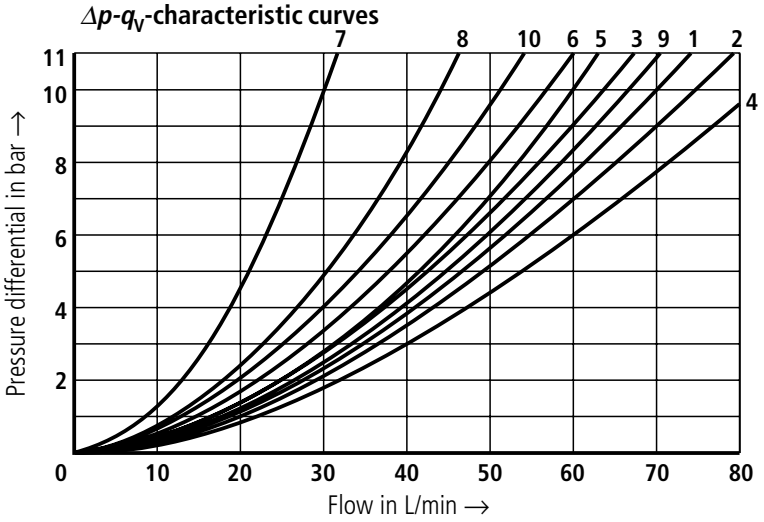
(other voltages on request)



AC solenoid		
Char. curve	**Solenoid voltage**	
19 to 28	W110	110 V, 60 Hz
	W230	230 V, 60 Hz

(other voltages on request)

Characteristic curves (measured at $\nu = 41$ mm²/s and $\vartheta = 50$ °C)



7 Symbol "R" in switched position A – B

8 Symbols "G" and "T" in mid position P – T

Symbols	Flow direction			
	P – A	P – B	A – T	B – T
A, B	3	3	–	–
C	1	1	3	1
D, Y	5	5	3	3
E	3	3	1	1
F	1	3	1	1
T	10	10	9	9
H	2	4	2	2
J, Q	1	1	2	1
L	3	3	4	9
M	2	4	3	3
P	3	1	1	1
R	5	5	4	–
V	1	2	1	1
W	1	1	2	2
U	3	3	9	4
G	6	6	9	9

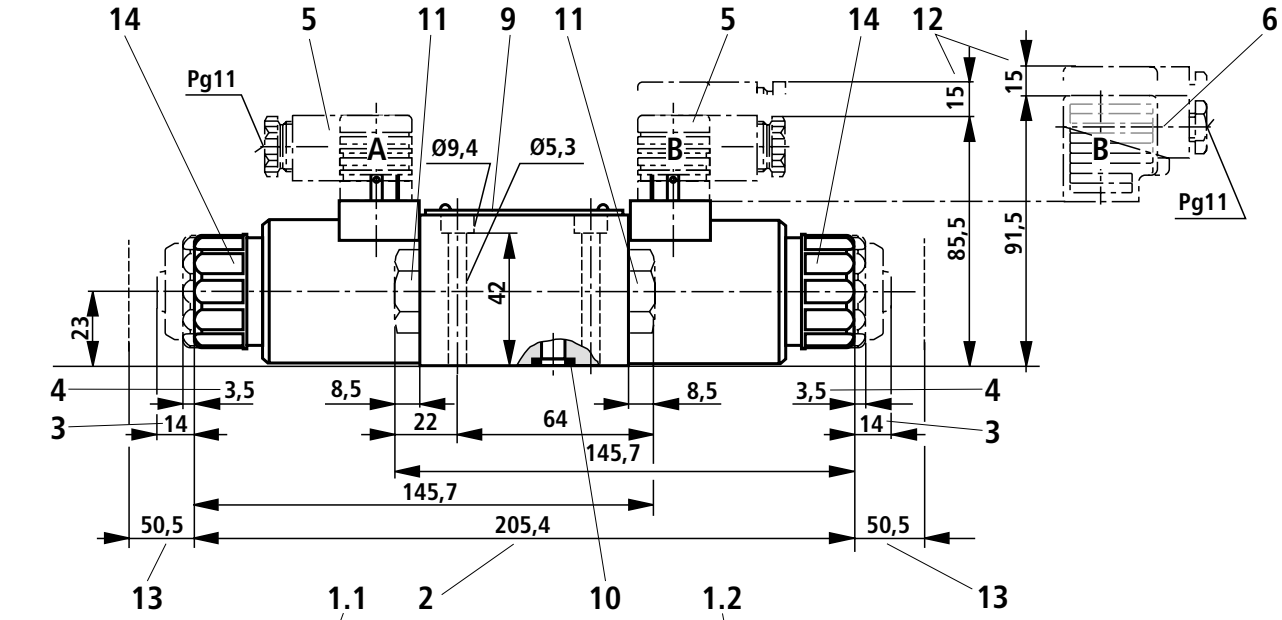
Preferred types (readily available)

Type	Material number
4WE 6 J6X/EG12N9K4	**00567496**
3WE 6 A6X/EG24N9K4	**00561180**
3WE 6 B6X/EG24N9K4	**00561270**
4WE 6 C6X/EG24N9K4	**00561272**
4WE 6 C6X/OFEG24N9K4	**00564107**
4WE 6 D6X/EG24N9K4	**00561274**
4WE 6 D6X/0FEG24N9K4	**00567512**
4WE 6 E6X/EG24N9K4	**00561278**
4WE 6 G6X/EG24N9K4	**00561282**
4WE 6 H6X/EG24N9K4	**00561286**
4WE 6 HA6X/EG24N9K4	**00549534**
4WE 6 J6X/EG24N9K4	**00561288**
4WE 6 M6X/EG24N9K4	**00577475**
4WE 6 Q6X/EG24N9K4	**00561292**
4WE 6 R6X/EG24N9K4	**00571012**
4WE 6 T6X/EG24N9K4	**00934414**
4WE 6 U6X/EG24N9K4	**00572785**
4WE 6 W6X/EG24N9K4	**00568233**
4WE 6 Y6X/EG24N9K4	**00561276**

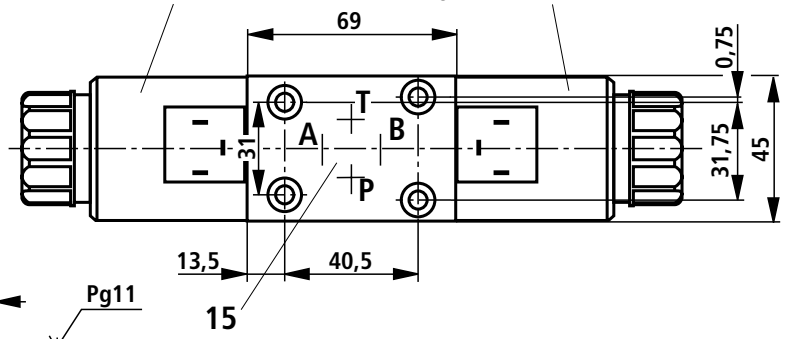
Type	Material number
4WE 6 D6X/EW110N9K4	**00551704**
4WE 6 D6X/OFEW110N9K4	**00552321**
4WE 6 E6X/EW110N9K4	**00558641**
4WE 6 J6X/EW110N9K4	**00551703**
4WE 6 A6X/EW230N9K4	**00915672**
4WE 6 B6X/EW230N9K4	**00915674**
4WE 6 C6X/EW230N9K4	**00913132**
4WE 6 D6X/EW230N9K4	**00909559**
4WE 6 D6X/OFEW230N9K4	**00915095**
4WE 6 E6X/EW230N9K4	**00912492**
4WE 6 G6X/EW230N9K4	**00912493**
4WE 6 H6X/EW230N9K4	**00912494**
4WE 6 HA6X/EW230N9K4	**00912495**
4WE 6 J6X/EW230N9K4	**00911762**
4WE 6 Y6X/EW230N9K4	**00909415**

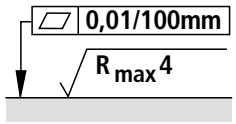
Preferred types and standard components are shown in the EPS price list (RDE 00 165).

Unit dimensions: valve with DC solenoid (Dimensions in mm)

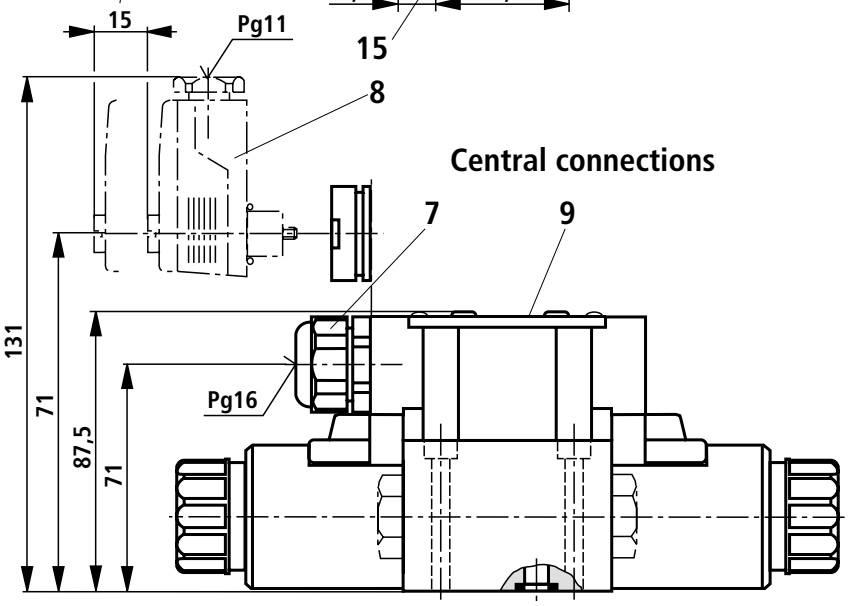


Individual connections





Required surface finish of mating piece



1.1 Solenoid "a" (plug-in connector colour grey)

1.2 Solenoid "b" (plug-in connector colour black)

2 Dim. for solenoid **with protected** hand override **"N9"** (standard)
– The hand override can only be actuated up to a tank pressure of approx. 50 bar

Avoid damage to hand override pin bore!

3 Dim. for solenoid **with** hand override **"N"**

4 Dim. for solenoid **without** hand override

5 Plug-in connector **without** circuitry to DIN 43 650 [1)]

6 Plug-in connector **with** circuitry to DIN 43 650 [1)]

7 Cable gland Pg 16 "DL"

8 Angled plug (colour red, must be ordered separately, Material no. 00005538)

9 Name plate

10 R-ring 9.81 x 1.5 x 1.78

11 Plug for valves with one solenoid

12 Space required to remove plug-in connector

13 Space required to remove coil

14 Securing nut, tightening torque M_A = 4 Nm

15 Porting pattern to DIN 24 340 form A, ISO 4401 and CETOP–RP 121 H

Subplates G 341/01 (G 1/4)
G 342/01 (G 3/8)
G 502/01 (G 1/2)

to catalogue sheet RE 45 052 and

Valve fixing screws
M5 x 50 DIN 912-10.9, M_A = 8.9 Nm, must be ordered separately.

Pin allocation for central connections:

With 1 solenoid:

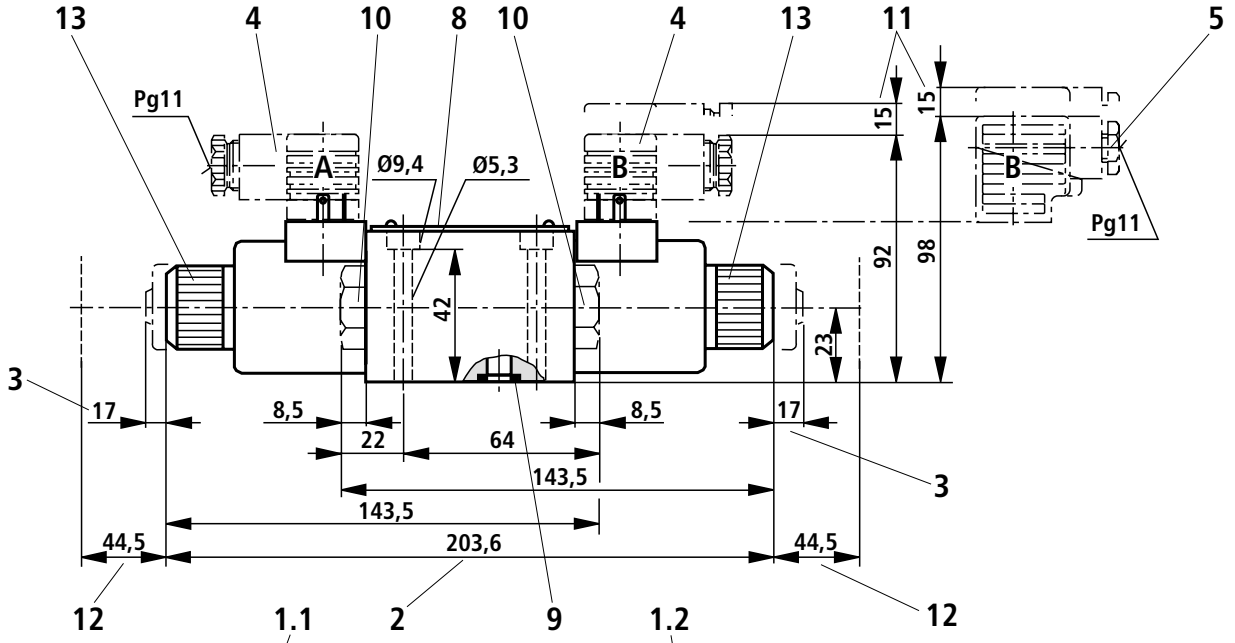
Solenoid always to terminals 1 and 2
Earth to terminal ⏚ PE

With 2 solenoids:

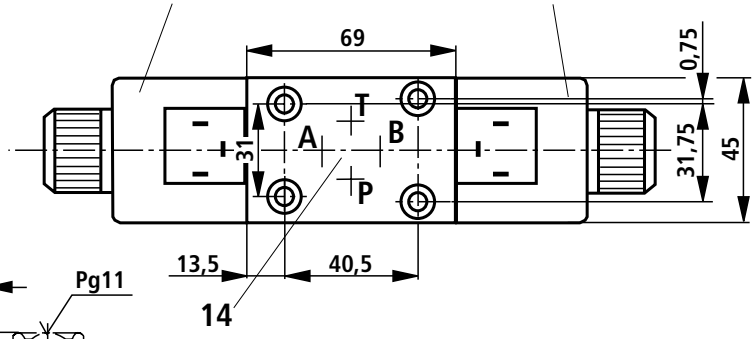
Solenoid "a" to terminals 1 and 2
Solenoid "b" to terminals 3 and 4
Earth to terminal ⏚ PE

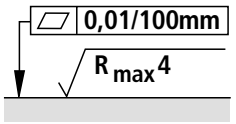
[1)] Must be ordered separately, see page 3.

Unit dimensions: valve with AC solenoid (Dimensions in mm)



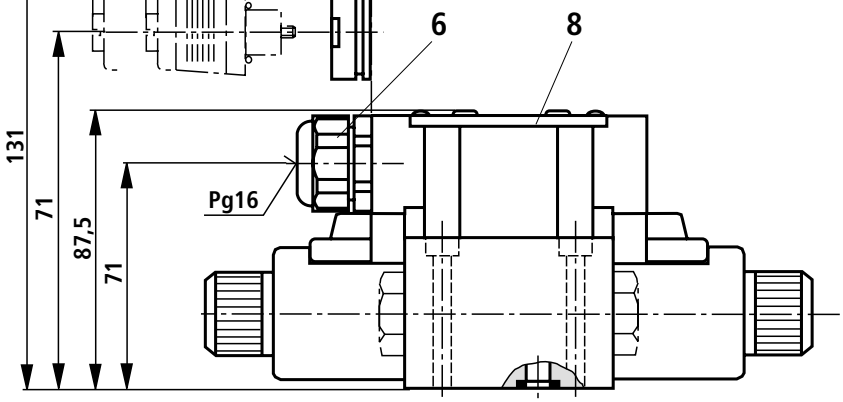
Individual connections





Required surface finish of mating piece

Central connections



Pin allocation for central connections:

With 1 solenoid:

Solenoid always to terminals 1 and 2
Earth to terminal ⏚ PE

With 2 solenoids:

Solenoid "a" to terminals 1 and 2
Solenoid "b" to terminals 3 and 4
Earth to terminal ⏚ PE

1.1 Solenoid "a" (plug-in connector colour grey)

1.2 Solenoid "b" (plug-in connector colour black)

2 Dim. for solenoid **with protected** hand override **"N9"** (standard) and **without** hand override
– The hand override can only be actuated up to a tank pressure of approx. 50 bar.

Avoid damage to the hand override pin bore!

3 Dim. for solenoid **with** hand override **"N"**

4 Plug-in connector **without** circuitry to DIN 43 650 [1)]

5 Plug-in connector **with** circuitry to DIN 43 650 [1)]

6 Cable gland Pg 16 "DL"

7 Angled plug (colour red, must be ordered separately, Material no. 00005538)

8 Name plate

9 R-ring 9.81 x 1.5 x 1.78

10 Plug for valves with one solenoid

11 Space required to remove plug-in connector

12 Space required to remove coil

13 Securing nut, tightening torque $M_A = 4$ Nm

14 Porting pattern to DIN 24 340 form A, ISO 4401 and CETOP–RP 121 H

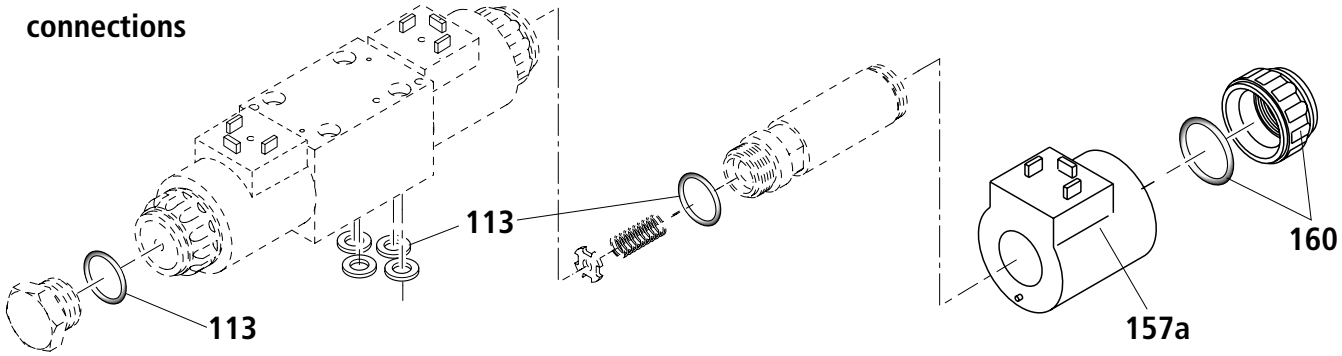
Subplates G 341/01 (G 1/4)
G 342/01 (G 3/8)
G 502/01 (G 1/2)

to catalogue sheet RE 45 052 and
Valve fixing screws
M5 x 50 DIN 912-10.9, $M_A = 8.9$ Nm, must be ordered separately.

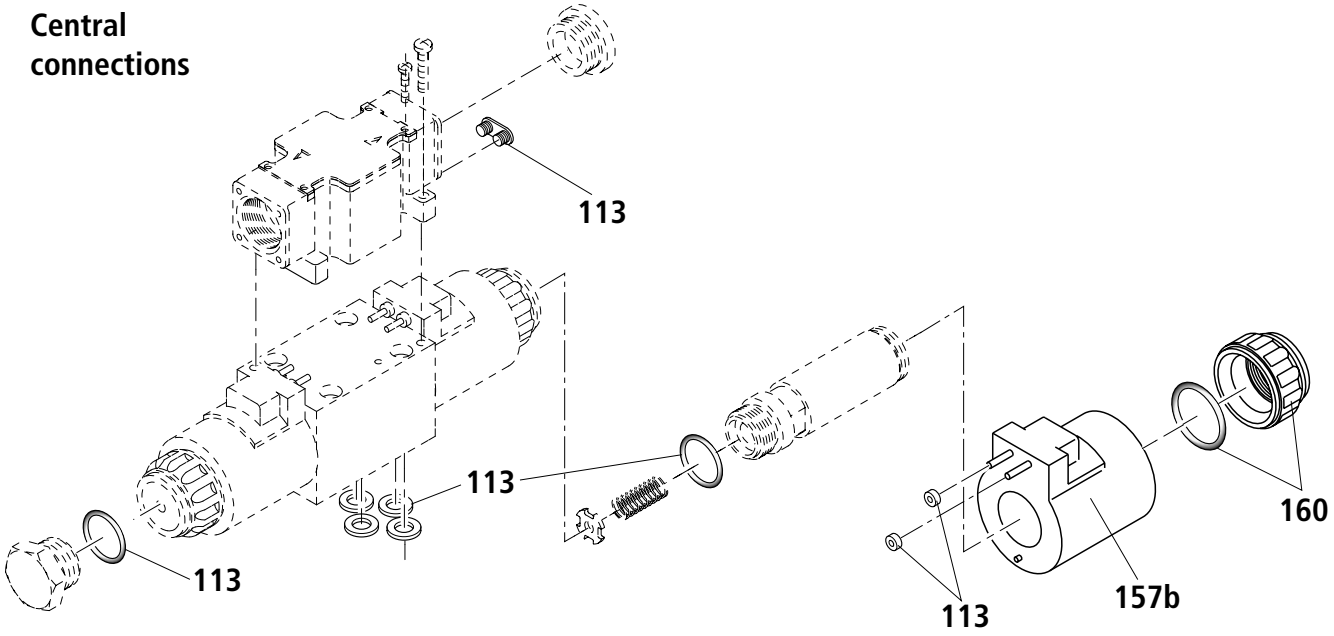
[1)] Must be ordered separately, see page 3.

Ordering details: available spare parts and seals

Individual connections



Central connections



Seal kit – valve: individual connection

Item	Seal material	Material number
113	NBR seals	00313162
	FKM seals	00313163

Seal kit – valve: central connection

Item	Seal material	Material number
113	NBR seal	00833687
	FKM seal	00833689

Spare parts – solenoid

Item	Description	DC		AC	
		Voltage	Material number	Voltage	Material number
157a	Coil for individual connections	12 V	00021388	110 V, 50/60 Hz	00020175
		24 V	00021389	230 V, 50/60 Hz	00071030
157b	Coil for central connections	12 V	00021462	110 V, 50/60 Hz	00021464
		24 V	00021463	230 V, 50/60 Hz	00071035
160	Nut: **with protected** hand override „N9“		00068604		00833831
	Nut: **with** hand override „N“		00227435		00833808
	Nut: **without** hand override		00227433		00833831

Notes

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